

Ömrüm Ceren GÜLER

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SUMMARY

Software Engineering graduate from an ABET-accredited program with 100% English instruction and a track record of academic excellence as a High School Valedictorian. Highly proficient in Python and FastAPI, with hands-on experience in building scalable backend architectures, predictive AI models, and computer vision tools during high-impact internships. Skilled in applying rigorous engineering methodologies such as Agile, COCOMO, and PERT/CPM to deliver efficient, data-driven solutions. Passionate about solving complex technical challenges and leveraging emerging technologies to drive innovation in large-scale software environments.

EDUCATION

Eastern Mediterranean University | B.S. in Software Engineering | Class of 2026 (3rd July)

- ABET Accredited, 100% English medium of instruction

Private Adana Ceyhan Final Anatolian High School | High School Diploma | Class of 2022

(Valedictorian, CGPA: 98/100)

PROFESSIONAL EXPERIENCE

Microsoft | AI Software Engineering Project Intern (AI Innovators) | August 2026 – September 2026

Istanbul, Turkey (4-Week Remote Internship)

- **Local RAG & Offline Inference Architecture:** Engineered an offline-first RAG pipeline using Azure AI Foundry and SQLite vector search, reducing query latency by 30% and achieving 1–3 second response times with zero network dependency.
- **Performance Optimization & Prompt Engineering:** Increased retrieval accuracy by 40% and eliminated hallucinations by implementing semantic chunking and strict prompt grounding for secure, edge-based LLM inference.
- **System Evaluation & Deployment:** Managed the complete SDLC process under the direct technical mentorship of Microsoft Turkey CSA Manager Barbaros Günay, executing functional system evaluations and GitHub-based code deployment.

RWS Group (TrainAI) | AI Data Specialist (Freelance / Project-Based) | May 2026 – Present

Maidenhead, England, United Kingdom (Remote)

- **AI Model Improvement:** Improving AI model assistant responses through various data annotation and data collection tasks to support the development of the next generation of artificial intelligence models.
- **Data Annotation & Tagging:** Performing data evaluations and tagging across multiple content types, including audio, video, and image annotation.
- **Content Evaluation:** Conducting pairwise comparisons and counting tasks to evaluate and improve the quality of AI-generated data.
- **Data Creation:** Creating clear, well-crafted written inputs and outputs specifically used for training and shaping AI models.

Çukurova University, IT Department | Software Engineering Intern | July 2025 – September 2025

Adana, Turkey (On-Site)

- **Backend & API Engineering:** Developed high-performance services using FastAPI and SQLAlchemy to automate data retrieval from SQL Server, reducing manual query overhead and ensuring 100% data integrity.
- **Predictive AI Services:** Integrated Facebook Prophet and Random Forest models into production APIs to forecast academic trends, delivering data-driven strategic insights for university leadership.
- **Computer Vision Automation:** Engineered a biometric compliance tool using MediaPipe and OpenCV to automate ID photo verification, resolving real-time latency issues in high-volume requests.
- **Data Intelligence Dashboards:** Deployed interactive web applications using Streamlit and Plotly Express, visualizing complex organizational hierarchies for non-technical stakeholders.
- **AI Productivity & Integration:** Leveraged advanced **Prompt Engineering** techniques to optimize software development workflows, reducing manual coding overhead and automating unit test generation. Expert in

designing structured prompts for LLMs to ensure consistent and reliable JSON outputs for seamless API integrations.

TECHNICAL PROJECTS

Foundry RAG Cloud (Live Cloud Deployment) | Independent Project | Aug 2026 – Present

- **System & REST Architecture:** Architected a production-minded Grounded RAG platform combining a standalone FastAPI/Uvicorn REST API backend with a Streamlit UI; implemented Clean Architecture (Ports & Adapters) to strictly decouple core logic from Azure OpenAI (text-embedding-3-small, gpt-4.1-mini).
- **Vector Storage & Resilience:** Built a custom SQLite cosine-similarity vector store with WAL journaling, atomic rollbacks (29 chunks indexed across 10 technical docs), NaN/Inf validation, and dynamic Azure 429 Retry-After exponential backoff.
- **DevOps, QA & Observability:** Implemented real-time per-query latency, confidence, and Cache HIT/MISS telemetry; enforced an 85% GitHub Actions CI coverage gate across 69 automated pytest cases with strict MyPy typing and non-root Docker containerization.
- **Tools Used:** Python 3.10+, FastAPI, Uvicorn, Streamlit, Azure OpenAI API, SQLite (Vector Engine), Docker, GitHub Actions (CI / YAML), Pytest, MyPy, Ruff, Clean Architecture.

Note: Full engineering portfolio of 10+ repositories available on my [GitHub](#) profile and here: [Portfolio](#)

TECHNICAL SKILLS & METHODOLOGIES

- **Languages & Databases:** Python (Advanced), Java, C, JavaScript (ES6+), R Language, SQL (PostgreSQL, SQL Server, MySQL, SQLite, MSSQL), SQLAlchemy, PyODBC, Node.js.
- **Frameworks & Web:** FastAPI, Uvicorn, Flask, RESTful APIs, Swagger (OpenAPI), MediaStream API, HTML5, CSS3, Streamlit, Pydantic v2, Plotly Express, ggplot2, Matplotlib, Seaborn, JSON, .env, httpx.
- **AI & Computer Vision:** OpenCV, MediaPipe, InsightFace (Embeddings & Cosine Similarity), PyTorch, Scikit-learn (K-Means, Random Forest), Pandas, NumPy, Prophet, Liveness Detection, AASIST (Graph Attention Networks), Deepfake Mitigation.
- **Software Management:** Agile (Scrum), Evolutionary SDLC, Effort Estimation (COCOMO), Resource Scheduling (PERT/CPM), Fishbone (Ishikawa) Diagrams, Pareto Analysis, QFD Matrix, Requirement Analysis, Testing Planning.
- **Engineering Practices:** Clean Architecture (Ports & Adapters), System Observability & Telemetry, Cloud Resilience & Bounded Retries, Atomic Transactions, Secure Coding Practices, UML Modeling, Root-Cause Analysis (RCA), Technical Troubleshooting, API Integration, System Architecture Design, Data Normalization, Unit/Integration Testing (pytest, pytest-cov), Type Checking (MyPy), Linting (Ruff, Flake8).
- **Ecosystem & Digital Media:** Digital Content Operations, Platform Ecosystems, Metadata Standards, Content Optimization, Digital Creator Technical Support, Media Lifecycle Awareness.
- **Tools & Standards:** Git/GitHub, CI & DevSecOps Pipelines (GitHub Actions, YAML, Bash), Docker, Docker Compose, MatLab, TinkerCad, Watchdog, MS Visio, MS Project (Gantt Charts), Modelio, Figma, GDPR & PCI-DSS Compliance, ABET Engineering Standards.
- **AI & Large Language Models:** Azure OpenAI, LLM Integration (OpenAI API, Anthropic, Llama), RAG (Retrieval-Augmented Generation), Local Vector Retrieval, Prompt Engineering, Function Calling, JSON Structured Outputs, AI-Assisted Development (GitHub Copilot, Cursor).

CERTIFICATIONS & ACHIEVEMENTS

- **Eastern Mediterranean Univ. (EMU):** GIT & GITHUB | Career & Coding | AI & Data | AI Summit'24 | Woman in Technology | Impact of Data Science & AI
- **Turkcell:** Lexathon'23 Legal Tech Innovation | Geleceği Yazanlar: Data Science & AI
- **IBM & Kodluyoruz:** Explore Emerging Tech, Agentic AI | Cyberstart (2024), AI4Future (June 2026)
- **Software Engineer Certifications:** HackerRank (May 2026) | micro1 (May 2026)
- **HP:** From Code to Creation: Mastering Game Programming (July 2026)
- **Additional: Research Report:** 5G Wireless & IoT Impact | **Languages:** English (Proficient), Turkish (Native)